

ESC101 20259 Design Brief Independent Assessment Tool

Requirements

1. **Shall** present an opportunity that can be addressed through a verifiable physical design concept
2. **Shall not** address any aspect of classroom design (i.e., classroom seating, storage, A/V, board visibility, etc.).
3. **Shall not** present preliminary or proposed engineering design(s) that address the opportunity
4. **Shall** be submitted as a single PDF file through Quercus.
5. **Should not** exceed 1800 words of text, inclusive of tables and exclusive of references, graphical elements, or appendices
6. **Shall** include a total word count at the top of the document.
7. **Shall** incorporate a title that allows a reader to understand the opportunity
8. **Shall** be in 11-pt or 12-pt font with 1-1.25 line spacing and 1.0-inch margins
9. **Shall** be completely anonymous with respect to your team; that is, it **must not** include any information (e.g. names, student numbers, identifiable photos, team identifier, etc.) that could be used to identify members of your team in the body of the Brief itself
10. **Shall** include relevant extracts from any used references in an Appendix titled “Source Extracts.”
11. **Shall** have a file name that briefly states the opportunity.

Evaluation Criteria

	Unacceptable	Satisfactory	Good	Outstanding
Quality of the opportunity • Reflection of the team's and primary stakeholders' interests and priorities • Represents a stakeholder(s) need(s) • Solvable by a team of first-year Engineering Science students • Representable and solvable in multiple meaningful ways • Cannot be solved by selection design	☐ Description of the opportunity lacks connection(s) with the stakeholders' stated interests, interpreted priorities and/or the team's expressed interests, priorities or experiences. Stated stakeholder needs are vague or not supported (e.g. missing observation, interviews/conversations, or research) Opportunity obviously requires a level of background knowledge or technical skills outside the experience of first-year EngSci students (e.g. circuitry, materials, chemistry, etc.) or requires no skill because it can be solved with an “off-the-shelf” solution.	☐ Description of the opportunity references stakeholders' stated interests and priorities and the team's interests and priorities, although the descriptions may be in broad, general terms Stakeholder needs are supported with observations, interviews/conversations, and/or research. Opportunity may require some skilling up in specific areas, but can be done in the available time period.	☐ Team's priorities are used to justify the opportunity as specific and meaningful for supportable reasons The stakeholders are explained and their needs included by considering multiple perspectives, using observations, interviews/conversations, and/or research. Opportunity offers multiple approaches to resolution rather than favouring a single “solution” to a problem. It invites divergence that will allow teams to find unique meaningful approaches.	☐ As per Good + Team's interests, priorities and experiences are incorporated with the stakeholder needs to create an opportunity that has focus, meaning, and sufficient detail to allow teams to move forward efficiently. The opportunity balances the need for challenge with the need for structure to create an excellent opportunity for a group of first-year EngSci students.
Correctness, appropriateness, and credibility of the Stakeholder Expectations • Models the opportunity and stakeholder expectations in engineering terms • Incorporates codes and standards, handbooks, DfXs, as necessary • Usable for Engineering Science students in their 1st term of study with limited time to complete their design activities • Internal consistency and consistency with remainder of the Brief	☐ Stakeholder Expectations missing critical components (e.g. need, goals, or objectives) or present but will not allow alternative designs to be evaluated (e.g. metrics are unconnected to solution aspect), and/or components are not credible (e.g. constraints come from contrived numbers). Objectives makes little or no reference to relevant standards, handbooks, or DfXs.	☐ Stakeholder Expectations will allow most alternative designs to be evaluated and compared, but may not be robust or fine-tuned enough to evaluate a broad range of potential designs Objectives founded in some credible basis (standards, guidelines, DfX definitions, or handbooks.)	☐ Stakeholder Expectations is well justified and supported by evidence, including research beyond the stakeholder statements into the context of the opportunity, research that supports engineering considerations of chosen DfXs; metrics and constraints/criteria are informed by relevant codes and/or standards Codes and standards are specific to the opportunity, as applicable, rather than general, or, if necessary, interpreted to connect them clearly to the opportunity.	☐ As per Good + There are clear connections between each level of the Stakeholder Expectations, with clear justification for each decomposition (eg. Goal to Objectives), such that a team can work with the whole set with minimal reframing. Uses of codes, standards, DfX are not only appropriate but are modified to be made usable for Engineering Science students in their 1st term of study who have limited time to complete their design activities.
Quality of reference designs • Selected reference designs demonstrate consideration of existing solution possibilities • Can be used to guide design activities	☐ Reference designs are missing; or reference designs resolve the opportunity, unless the opportunity is totally over-constrained.	☐ Reference designs are present but only used to show unsatisfactory design alternatives.	☐ Reference designs are used to show what prior work has been done toward a solution or in related areas, including both positive and negative features of a reference.	☐ As per Good + reference designs are used to demonstrate aspects of the opportunity that are not easily shown otherwise, such as design of specific features
The quality and credibility of your engineering arguments • Use data gathered from observation and research • Structured with an emphasis on reason	☐ An over-reliance on opinion, description, or "it's obvious" statements. A lack of evidence throughout. Range of evidence used is limited, with over-reliance on sources that are lacking in one or more of currency, relevance, authority, accuracy, or compatible purpose.	☐ Claims appear credible at first glance, with most supported by evidence. A range of resources are used (textbooks, handbooks, reviews, direct testing, interviews...), with t. Some evidence may be inappropriate or weak for the claim being made.	☐ Claims are supported by credible evidence throughout. Interpretive claims are supported by more substantial evidence and use more deliberate justification.	☐ As per Good + resources are used credibly, with qualifiers and triangulation regularly applied. Acknowledgements/analysis of possible {weaknesses, risk, error, omissions} are provided as needed.
Quality of the document as a design brief • Allows reader to find information quickly • Uses structure to aid readability • Structure responds to content	☐ Structure used is inappropriate to the genre. Introduction lacks some elements. Conclusion lacking or missing. Arrangement of information (including sub-headings) within the report is mainly inductive; it may be illogical or confusing. Paragraphs lack clear claims or support for claims.	☐ Structure is appropriate to the genre and attempts to use a deductive arrangement of information. Introduction contains all elements. Conclusion present. Arguments and headings follow a logical structure. Structural elements such as bullet lists or tables are present, but may be poorly used or over-used.	☐ An overall deductive arrangement of information is used consistently and effectively. Introduction is efficient and clear to establish purpose. Conclusion is effective. Headings and sub-headings provide a clear structure for a reader. Structural elements are used appropriately and aid in clarity and understanding.	☐ As per Good + the document guides readers through the report structurally and conceptually. Readers can understand the central claims of the document by skimming the sub-headings and topic sentences.
The coherence and clarity of your written and visual communication • Visual and text are chosen and integrated to clearly communicate meaning • Uses specific language of engineering and design	☐ No visuals present or are all relegated to the appendix. Visuals lack captions or visuals are illegible or irrelevant and thus not useful to enhance the document's message. Language errors and/or issues with editing and proofreading significantly detract from the readability and professionalism of the report.	☐ Visuals are present in the document (and captioned), but may not be well used in text. Visuals may be lacking in areas where they would aid in a reader's understanding. Writing is clear, but may be repetitive or show a disruptive preference for flourish over concision.	☐ Visuals are integrated with text and meaningfully discussed. Choice of visuals improves understanding and clarity. Writing is clear, concise, and demonstrates awareness of specific engineering and design language. Coherence is maintained throughout paragraphs and arguments.	☐ As per Good + visuals include appropriate guidance for the reader. Writing style and structure demonstrates nuanced control of language through appropriate use of qualifiers in arguments, clear and concise expression, and precision in word choices.